

Xieheng Wang

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Email: xiehengw@gmail.com Personal Website: <https://mariana2000.github.io/>

Research interests Statistical methods & applications to big data, Biomedical data science, Statistical genetics and genomics, Dimension reduction, Transfer learning

Education **Tsinghua University** **Beijing, China**

B.Eng. in Measurement, Control Technology and Instruments GPA: 3.69/4.00

Sept. 2019 – Jul. 2023

Minor in Statistics GPA: 3.92/4.00

Mar. 2021 – Jul. 2023

University of Minnesota, Twin Cities **MN, USA**

Ph.D. in Statistics GPA: 3.86/4.00

Aug. 2023 – Present

Research experience **Causal Inference between Multi-Omics traits and Alzheimer's Disease**

Mentor: Wei Pan (University of Minnesota) *Aug. 2023 – July. 2024*

- Applied 2ScML (2-stage constraint maximum likelihood) to OmicsPred pre-trained models and AD GWAS data (such as IGAP and EADB) to identify (putative) causal relationships between multi-omics traits and Alzheimer's Disease.

Relationships and Shared Genetic Influences between T2D Biomarkers and Human Brain Health

Mentor: Bingxin Zhao (University of Pennsylvania) *Apr. 2022 – Dec. 2022*

- Explored the association between Type 2 diabetes and human brain from individual level (polygenic risk score analysis) and summary level (LD Score Regression and Local Analysis of [co]Variant Association).

Pure Noise Enables Privacy-preserving Federated Learning with Non-iid Data

Mentor: Huaiyu Dai (North Carolina State University) *July 2022 – Aug. 2022*

- Applied OPeN (Oversampling with Pure Noise Image) in the framework of federated learning to enhance performance and privacy.

Feature-based Classification on MQTT Data

Mentor: Jingyi Zhang (Tsinghua University) *Oct. 2021 – Jun. 2023*

- Applied POTD (principal optimal transport direction) on MQTT-IoT-IDS2020 dataset to extract features distinguishing different attacks.
- Proposed a new ensemble method of multiclass classification based on SDR subspace using POTD.

Design and Construction of Vision-based Quadrotor Navigation Control System

Mentor: Kaichun Zhao (Tsinghua University) Sept. 2020 – Dec. 2021

- Applied some visual SLAM methods like SuperPoint on real image data obtained by drone in the environment of smoke.

Publications

J. Liu*, Y. Tang*, H. Zhao, **X. Wang**, F. Li, and J. Zhang, “CPS Attack Detection under Limited Local Information in Cyber Security: A Multi-node Multi-class Classification Ensemble Approach”, *ACM Transactions on Sensor Networks (TOSN)*, 2024. [\[pdf\]](#)

Teaching experience

Teaching Assistant, Tsinghua University

Orienteering Sports

Spring 2022

Teaching Assistant, University of Minnesota, Twins Cities

STAT 4101: Theory of Statistics I

Fall 2024

STAT 5021: Statistical Analysis

Spring 2025

STAT 5601: Nonparametric Methods

Spring 2025

STAT 4051: Statistical Machine Learning I

Summer 2025

Graduate Instructor, University of Minnesota, Twin Cities

STAT 4101: Theory of Statistics I

Fall 2025

Honors and scholarships

University of Minnesota, Twin Cities

Statistics Alumni Fellowship (Highest honor at School of Statistics) 2025-2026

School of Statistics First Year Scholarship 2023

Tsinghua University

Sports Excellence Scholarship, Tsinghua University 2022

Academic Excellence Scholarship, Tsinghua University 2021

Sports Excellence Scholarship, Tsinghua University 2021

Academic Excellence Scholarship, Tsinghua University 2020

Sports Excellence Scholarship, Tsinghua University 2020

Skills

Programming

R, Python, PyTorch, MATLAB, C/C++

Tools

LaTeX, Linux, SPSS, Git, OCAD

Languages

English (TOEFL:102, CET-6:630), Mandarin (native)

Leadership&Activities	Vice president of Tsinghua University Student Orienteering Association	<i>Sept. 2022 - Jun. 2023</i>
	Member of Tsinghua Orienteering Team	<i>Oct. 2019 - Jun. 2023</i>
	Volunteer, mapper and member of Minnesota Orienteering Club (MNOC)	<i>Sept. 2023 - Present</i>
	Volunteer of STATGEN 2025, Minneapolis	<i>May 2025</i>